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## Professional Summary

I build scientific software that transforms biological data into reproducible quantitative measurements.

Throughout my career I have worked at the intersection of biology, quantitative analysis, and software engineering. Rather than developing one-off analysis scripts for individual publications, I design reusable software platforms that enable researchers to analyze imaging and electrophysiology data through reproducible computational workflows. My work combines extensive scientific domain expertise with modern software engineering practices to make data analysis easy to perform, reproducible, and shareable.

## Scientific Domains

My scientific domain expertise includes neuroscience, vascular biology, and cardiac physiology. I have decades of experience at the bench designing experiments to test new hypotheses, performing experiments with precision, interpreting results, and publishing the findings. I have also built custom microscopy and electrophysiology rigs, developed software for real-time data acquisition and visualization, and supported researchers using these systems. This experience connects experimental design and data acquisition with the quantitative analysis supported by my software.

This expertise developed along a career trajectory from studying neurons to imaging brain vasculature and later working in cardiac physiology. This trajectory showed me that fields organized around different organs and research questions often rely on shared experimental technologies and face common analysis problems. Regardless of the specimen or scientific question, imaging experiments create similar practical needs for organizing and visualizing data, making quantitative measurements, and producing reproducible results. My direct experimental work has centered on microscopy and electrophysiology, but I design software that can be extended with domain experts to support new experimental systems and scientific questions.

## Research Software Engineering

I design reusable scientific software rather than bespoke analysis scripts. I begin with the scientific question and experimental workflow, then develop modular computational backends that can support multiple applications and analyses. My work combines documented interfaces, automated testing, interactive desktop and web applications, and open-source development.

I design each analysis system as one continuous software environment built around a shared computational backend with graphical and scripting interfaces. The same software remains with a project while data are acquired, during offline analysis, and at publication. At every stage, the interfaces use the same data model and analytical methods, preserving reproducibility without duplicating calculations. Publication is designed into the architecture rather than added after analysis. This makes data sharing part of the analysis workflow rather than a separate task created at the end of a project.

My longer-term goal is to publish raw data and completed analyses with the same software used in the laboratory so readers can inspect how figures were produced, examine reported results, repeat the analysis, and extend it to address new scientific questions. I am extending these systems to prepare data and completed analyses for community standards and appropriate repositories selected according to the scientific domain and data type.

Scientific software has the greatest impact when it becomes reusable research infrastructure. Reusable software platforms reduce duplicated effort, improve reproducibility, and allow laboratories to build upon stable computational foundations rather than repeatedly developing new analysis scripts for each project. By combining sustainable software engineering with open-source development, research infrastructure continues to support scientific discovery long after individual publications, grants, and laboratory personnel have changed. This long-term perspective enables software to become a lasting scientific resource rather than a temporary research product.

## Research Software Platforms

***CloudScope*** · Live Web App · Documentation

Microscopy image analysis often depends on disconnected tools and manual processing. CloudScope provides desktop and web interfaces for data management, visualization, and quantitative analysis. Its interfaces use AcqStore as their shared computational backend. Analyses for blood-flow velocity, vessel diameter, heart rate, and related signals are implemented in AcqStore. They are available through CloudScope without duplicating the underlying calculations. New AcqStore file loaders, analyses, and visualization tools can also be incorporated into CloudScope as the backend evolves.

***AcqStore*** · Documentation · Source Code

AcqStore is a general-purpose Python toolbox and computational backend for loading, managing, visualizing, and analyzing imaging data. Its documented API supplies data, visualization primitives, and analysis methods to CloudScope and scripting workflows.

Raw imaging data come from many sources, including proprietary microscope file formats. AcqStore loads these data while retaining the experimental metadata and physical units required for quantitative analysis. Its extensible file-loader plugin system allows new formats to be supported without rewriting applications built on AcqStore.

Plugins extend the platform with capabilities for analysis and cloud sharing, while an extensible export system produces structured tabular results. Desktop, web, and scripted workflows therefore share the same data model and analysis implementation, improving reproducibility while allowing new capabilities to be added over time.

Lazy loading enables collections containing hundreds or thousands of files to be browsed without loading the complete dataset into memory. Support for OME-Zarr and NGFF file formats provides scalable data access, sharing, and cloud deployment.

***SanPy*** · Documentation · Source Code

Electrophysiology analysis frequently depends on manual measurements or laboratory-specific scripts. Manual measurements are difficult to standardize and reproduce across researchers and experiments. SanPy provides automated action-potential detection, quantitative measurements, interactive visualization, a plugin architecture, and a documented Python API.

The freely available desktop application supports real-time use during experiments, offline analysis, and inspection of published analyzed datasets. Its GUI and scripting workflows use the same internal computational backend. This ensures that interactive and automated workflows calculate measurements with the same implementation, improving reproducibility while allowing analyses to be extended and automated.

***MapManager*** · Live Web App · Documentation

Longitudinal microscopy studies require researchers to organize, visualize, and analyze datasets collected across multiple imaging sessions potentially spanning weeks to months. MapManager provides an open-source platform for reproducible annotation, visualization, and quantitative analysis of dendritic spine dynamics over time.

MapManager is optimized to manage annotation collections that may contain tens of thousands of items. Its graphical interface supports interactive review and curation, including identifying and correcting false-positive and false-negative annotations.

***PiE*** · Documentation · Source Code

Behavioral neuroscience experiments often require custom hardware control, data acquisition, and analysis software assembled from independent components. Experimenter presence can also interfere with sensitive behaviors and confound measurements. PiE integrates experiment control, automated acquisition, data management, video streaming, and quantitative analysis. Its web interface allows experiments to be controlled and monitored remotely, reducing experimenter interference while supporting reproducible behavioral research.

***Brightest Path Tracing*** · Documentation · Source Code

Python lacked a reusable, documented library for tracing brightest paths through n-dimensional scientific images. Brightest Path fills this gap by providing a tested, installable Python package with a well-documented API, enabling scientific software projects to incorporate robust path-tracing algorithms without reimplementing them.

## Technical Skills

**Programming and scripting:** Python, C/C++, Igor Pro, Bash, zsh

**Scientific computing:** NumPy, SciPy, pandas, PyTorch, scikit-image, multiprocessing, multithreading

**User interfaces and visualization:** PyQt, pyqtgraph, NiceGUI, napari, pywebview, Plotly, Matplotlib

**Scientific data and formats:** HDF5, Zarr, OME-Zarr, NGFF, s3fs, lazy loading, image pyramids

**Web applications and APIs:** HTML, JavaScript, WebAssembly, Pyodide, FastAPI, uvicorn, Pydantic, HTTP and JSON APIs, OpenAPI, httpx, thin-client architecture

**Software engineering, testing, and documentation:** Git, GitHub, pytest, GitHub Actions, uv, MkDocs, documented Python APIs, Google-style docstrings, end-user and developer documentation

**Deployment and infrastructure:** Docker, Docker Compose, PyInstaller, macOS and Windows desktop applications, Linux-based development and continuous integration

**Scientific analysis:** quantitative microscopy, electrophysiology analysis, time-series analysis, image segmentation, ROI-based analysis, longitudinal annotation analysis, brightest-path tracing

**Scientific instrumentation and acquisition:** laser-scanning microscopy, custom microscopy and electrophysiology acquisition systems, whole-cell current-clamp electrophysiology, real-time data acquisition and visualization, Arduino microcontrollers, remote experiment control and video monitoring

## Leadership and Mentorship

My software projects have supported collaborative research involving faculty, postdoctoral scholars, graduate students, and undergraduate researchers. I enjoy mentoring scientists in quantitative analysis, software design, and computational methods while developing software that enables research groups to become more productive and self-sufficient.

I involve researchers throughout software development, from initial design and implementation through testing and critical feedback. Scientific software is a living resource. Continued feedback from its users keeps it useful as experimental workflows and scientific questions evolve.

I value the day-to-day work that makes shared research infrastructure useful. This includes working directly with researchers, troubleshooting datasets and analysis pipelines, improving documentation, and providing practical training. Recurring support problems often reveal needs shared across laboratories. I use that experience to improve the software and create solutions that can be reused.

## Teaching and Scientific Training

My teaching connects scientific instrumentation with quantitative analysis and interpretation. Through lectures, laboratory instruction, and research training, I have taught the optical physics of laser-scanning microscopy, image formation, the physical limits of light microscopy, and signal-detection principles relevant to imaging and electrophysiology. This training helps researchers understand not only how to operate an instrument, but also what it measures, where uncertainty enters, and how acquisition choices affect analysis and interpretation.

I also designed and taught an undergraduate Internet of Things (IoT) course as instructor of record for three years. The course combined IoT concepts with hands-on circuit building with Arduino microcontrollers. Lectures and final projects examined current medical applications of IoT, including wearable technologies and real-time, remote data acquisition. We also considered how large longitudinal datasets could support new scientific questions.

I have also mentored computer science and biophysical engineering graduate students and managed full-time employees with computer science backgrounds. I helped them apply their expertise in mathematics, physics, and software engineering to biological questions and experimental workflows. This ability to build a shared language across disciplines is important for supporting imaging-core and analysis-core users.

## Employment

2019-2026 Assistant Professor
Department of Physiology and Membrane Biology
School of Medicine
Affiliations: Computer Science Graduate Group, Neuroscience Graduate Group, DataLab
University of California – Davis, CA

2015-2019 Research Associate
Solomon H. Snyder Department of Neuroscience
The Johns Hopkins University School of Medicine, Baltimore, MD

2009-2015 Postdoctoral Fellow
Solomon H. Snyder Department of Neuroscience
The Johns Hopkins University School of Medicine, Baltimore, MD

Advisor: David J Linden, Ph.D.

2005–2009 Postdoctoral Fellow
L'Unité de Neurobiologie des canaux Ioniques et de la Synapse
Université Aix Marseille, Marseille FRANCE
Advisor: Dominique Debanne, Ph.D.

1993–1998 Scientific Software Engineer
Roswell Park Cancer Institute
University at Buffalo, Buffalo, NY
Advisor: Kenneth Manly, Ph.D.

**Education**

1998–2004 Brandeis University, Ph.D. Neuroscience
Advisor: Gina G Turrigiano, Ph.D.

1996–1998 University of Pennsylvania, Graduate studies
School of Engineering, Department of Computer and Information Science

1988–1992 State University of New York at Buffalo
B.A. Computer Science, Joint degree Media Studies

**Funding**

2025-2026 NIH R01, NHLBI (5R01HL168874-02)
Metabolic Control of Cardiac Pacemaking
Role: PI (with Santana LF and Clancy CE)

2021-2025 NIH R01, BRAIN Initiative, NIMH (1R01MH123206-01A1)
Map Manager: Software for image volume time-series analysis and sharing.
Role: PI

2022-2023 Chan Zuckerberg Initiative (CZI).
Napari plugin development.
Role: PI

2022-2023 NHLBI, Supplement to RO1 (3R01HL152681-02S1).
Role: Personnel
PI: Santana LF

2016-2020 American Heart Association, Scientist Development Grant
Role: PI

**Professional Service**

2024/2025 NIH Research Software Engineer (RSE) Award, Study Section
2025 NSF Graduate Research Fellowship Program (GRFP), Review Panel

2015-2026 Manuscript Review
Review Editor, Frontiers in Physiology (Vascular Physiology)
Review Editor, Frontiers in Neuroscience (Neuroimaging – Brain Imaging Methods)
Journal of Open Source Software (JOSS), eNeuro, eLife, J Physiology, J Neurophysiology

**<u>Selected Publications · PubMed for full list</u>**


Benedict J, **Cudmore RH**, Oden D, Spruell A, Linden DJ (2025) The Lateral Habenula Is Necessary for Maternal Behavior in the Naturally Parturient Primiparous Mouse Dam. eNeuro 13:12(1). PubMed: 39689968.

Guarina L, Le JT, Griffith TN, Santana LF, **Cudmore RH** (2024) SanPy: Software for the analysis and visualization of whole-cell current-clamp recordings. Biophys J 123:759–769. PubMed: 38419330.

Jha V, **Cudmore RH** (2024) Brightest path tracing: A Python package to trace the brightest path in 2D and 3D images. bioRxiv:2023.07.16.549233. PubMed: 37503184.

Lopez-Ortega E, Choi JY, Hong I, Roth RH, **Cudmore RH**, Huganir RL (2024) Stimulus-dependent synaptic plasticity underlies neuronal circuitry refinement in the mouse primary visual cortex. Cell Rep 43(4):113966. PubMed: 38507408.

Benedict J, **Cudmore RH** (2023) PiE: an open-source pipeline for home cage behavioral analysis. Front Neurosci 17:1222644. PubMed: 37583418.

Grainger N, Guarina L, **Cudmore RH***, Santana LF* (2021) The Organization of the Sinoatrial Node Microvasculature Varies Regionally to Match Local Myocyte Excitability. Function 2:zqab031. PubMed: 34250490.

* Equal contribution, corresponding author

Li Y, Ding R, Wang F, Guo C, Liu A, Wei L, Yuan S, Chen F, Hou S, Ma Z, Zhang Y, **Cudmore RH**, Wang X, Shen H (2020) Transient ischemia-reperfusion induces cortical hyperactivity and AMPAR trafficking in the somatosensory cortex. Aging 12:4299–4321. PubMed: 32155129.

Roth RH, **Cudmore RH**, Tan HL, Hong I, Zhang Y, Huganir RL (2020) Cortical Synaptic AMPA Receptor Plasticity during Motor Learning. Neuron 105:895-908.e5. PubMed: 31901303.

Tan HL, Roth RH, Graves AR, **Cudmore RH**, Huganir RL (2020) Lamina-specific AMPA receptor dynamics following visual deprivation in vivo. Elife 9:e52420. PubMed: 32125273.

Ye Z, **Cudmore RH**, Linden DJ (2019) Estrogen-Dependent Functional Spine Dynamics in Neocortical Pyramidal Neurons of the Mouse. J Neurosci 39:4874–4888. PubMed: 30992373.

**Cudmore RH**, Dougherty SE, Linden DJ (2017) Cerebral vascular structure in the motor cortex of adult mice is stable and is not altered by voluntary exercise. J Cereb Blood Flow Metab 37(12):3725-3743. PubMed: 28059584.

Jin Y, Dougherty SE, Wood K, Sun L, **Cudmore RH**, Abdallah A, Pletnikov M, Hashemi P, Linden DJ (2016) Regrowth of Serotonin Axons in the Adult Mouse Brain Following Injury. Neuron 91(4):748-62. PubMed: 27499084.

Zhang Y*, **Cudmore RH***, Lin D, Linden DJ, Huganir RL (2015) Visualization of NMDA receptor– dependent AMPA receptor synaptic plasticity in vivo. Nature Neuroscience 18(3):402-7. PubMed: 25643295.

* Equal contribution

**Selected Conference Abstracts**

**Cudmore RH**, (2024) Open-source software for the analysis of laser scanning kymographs. Society For Neuroscience Abstract, Chicago. PSTR493.01.

**Cudmore RH** and Zawadzki RJ (2024) An image analysis toolbox for 3D vascular tracing and network topology. Proceedings Volume 12830, Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXVIII; 128300B. San Francisco. https://doi.org/10.1117/12.3001224

**Cudmore RH**, Guarina L, Le JT, Griffith TN, Santana LF (2023) Sanpy: an open-source whole-cell electrophysiology analysis pipeline. Society For Neuroscience Abstract. Washington. PSTR446.17.

Abunijem S, Le JT, **Cudmore RH** (2023) Map Manager: Software to annotate and analyze image volume time-series. Basic Sciences Symposium Celebrating Hispanic Heritage Month, UC Davis.

Manning D, **Cudmore RH**, Trimmer J, Santana LS (2023) KV2.1 is critical for vascular smooth muscle relaxation and neurovascular blood flow. Basic Sciences Symposium Celebrating Hispanic Heritage Month, UC Davis.

**Cudmore RH** (2018) Map Manager: Software to annotate and analyze image volume time-series. Society For Neuroscience Abstract. 254.17.

Roth RH, **Cudmore RH**, Tan HL, Zhan Y, Huganir RL (2018) Cortex-wide synaptic AMPA receptor plasticity during motor learning. Society For Neuroscience Abstract. 288.02.

Tan HL, Roth RH, Graves AR, **Cudmore RH**, Huganir RL (2018) Distance-dependent AMPA receptor dynamics following visual deprivation in vivo. Society For Neuroscience Abstract. 288.06